

```
In [43]: import pandas as pd
import matplotlib.pyplot as plt
data={
'Date':['21-8-2025', '22-8-2015', '23-8-2005', '24-8-2006'],
'Product': ['Laptop', 'Mouse', 'Keyboard', None],
'Sale': [1200, 25, 75, None],
'Quantity': [10, 50, 30, None],
'Region': ['Chennai', 'Madurai', 'Trichy', None]
}
df=pd.DataFrame(data)
df
```

Out[43]:

	Date	Product	Sale	Quantity	Region
0	21-8-2025	Laptop	1200.0	10.0	Chennai
1	22-8-2015	Mouse	25.0	50.0	Madurai
2	23-8-2005	Keyboard	75.0	30.0	Trichy
3	24-8-2006	None	NaN	NaN	None

```
In [44]: df.loc[3, 'Sale']=df['Sale'].mean()
df
```

Out[44]:

	Date	Product	Sale	Quantity	Region
0	21-8-2025	Laptop	1200.000000	10.0	Chennai
1	22-8-2015	Mouse	25.000000	50.0	Madurai
2	23-8-2005	Keyboard	75.000000	30.0	Trichy
3	24-8-2006	None	433.333333	NaN	None

```
In [45]: df.drop(index=3,axis=1,inplace=True)
df
```

Out[45]:

	Date	Product	Sale	Quantity	Region
0	21-8-2025	Laptop	1200.0	10.0	Chennai
1	22-8-2015	Mouse	25.0	50.0	Madurai
2	23-8-2005	Keyboard	75.0	30.0	Trichy

```
In [46]: df.describe()
df
```

Out[46]:

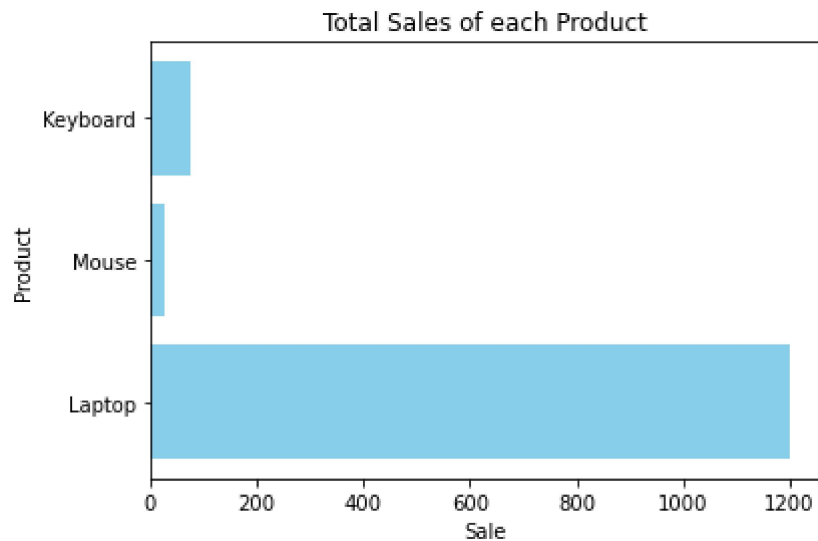
	Date	Product	Sale	Quantity	Region
0	21-8-2025	Laptop	1200.0	10.0	Chennai
1	22-8-2015	Mouse	25.0	50.0	Madurai
2	23-8-2005	Keyboard	75.0	30.0	Trichy

```
In [47]: df.groupby('Product').agg(  
    Total_Sales=('Sale', 'sum'),  
    Total_Quantity=('Quantity', 'sum')  
)
```

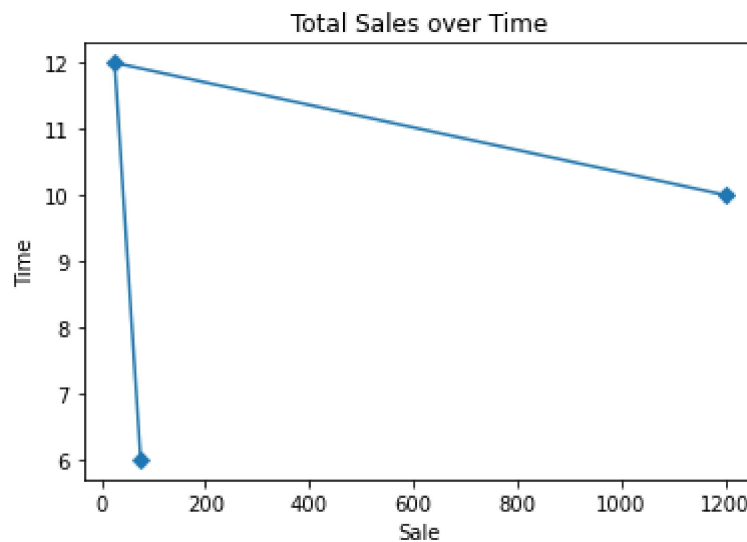
Out[47]:

	Total_Sales	Total_Quantity
Product		
Keyboard	75.0	30.0
Laptop	1200.0	10.0
Mouse	25.0	50.0

```
In [48]: df.to_csv("sales_data.csv",index=False)  
plt.barh(df["Product"],df["Sale"],color='skyblue')  
plt.title("Total Sales of each Product")  
plt.xlabel("Sale")  
plt.ylabel("Product")  
plt.show()
```



```
In [53]: df.rename(columns={'Date': 'DateTime'})
plt.plot(df['Sale'], df['DateTime'], marker='D')
plt.title('Total Sales over Time')
plt.xlabel('Sale')
plt.ylabel('Time')
plt.show()
```

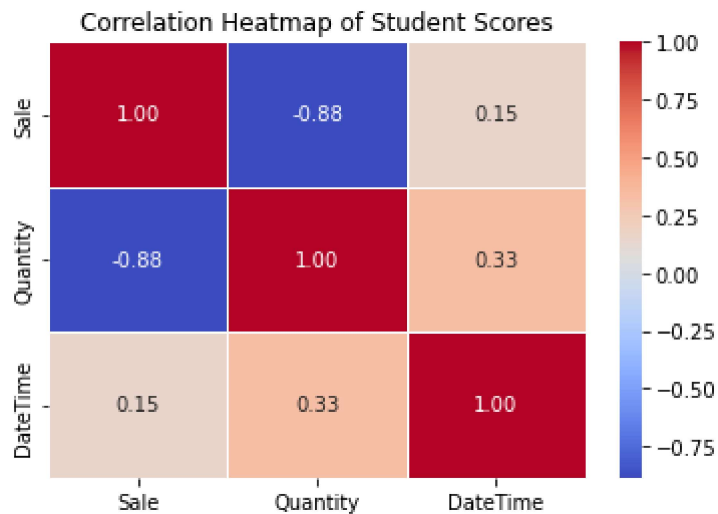


```
In [57]: pivot_table_multi_agg = pd.pivot_table(df, values=['Region', 'Product'], index='Sale')
print("\nPivot Table")
print(pivot_table_multi_agg)
```

Pivot Table

	Product	Region
Sale		
25.0	Mouse	Madurai
75.0	Keyboard	Trichy
1200.0	Laptop	Chennai

```
In [64]: g=df.corr()
sns.heatmap(g, annot=True, cmap='coolwarm', fmt=".2f",linewidths=.5)
plt.title('Correlation Heatmap of Student Scores')
plt.show()
```

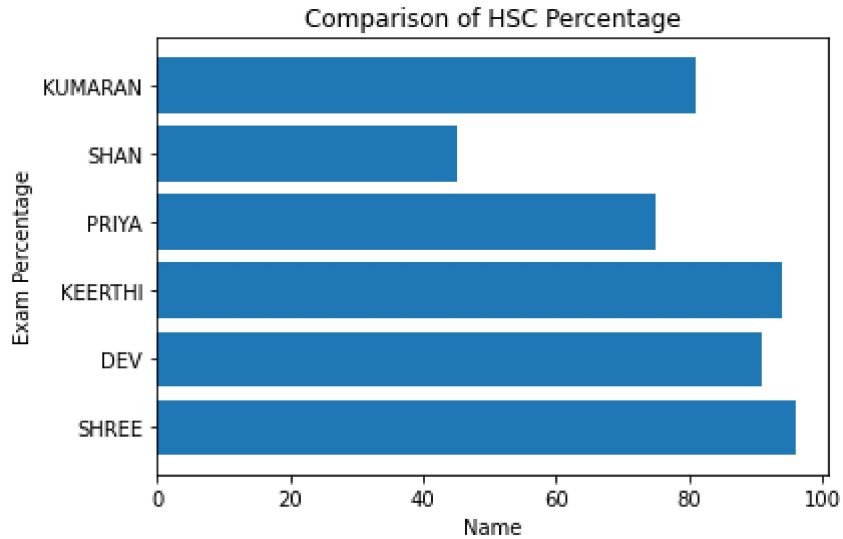


```
In [67]: import pandas as pd
import matplotlib.pyplot as plt
data={
'Name':['SHREE', 'DEV', 'KEERTHI', 'PRIYA', 'SHAN', 'KUMARAN'],
'Percentage': [96, 91, 94, 75, 45, 81]
}
df=pd.DataFrame(data)
df
```

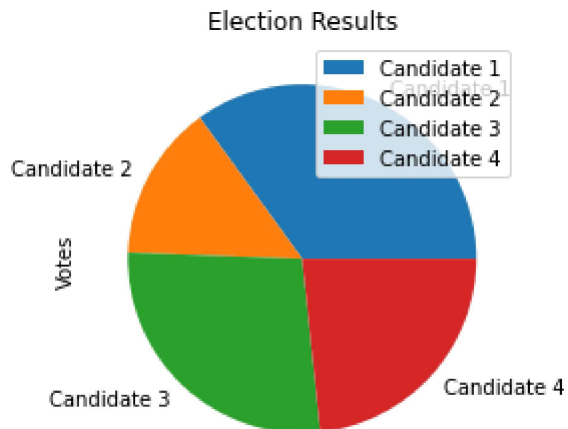
Out[67]:

	Name	Percentage
0	SHREE	96
1	DEV	91
2	KEERTHI	94
3	PRIYA	75
4	SHAN	45
5	KUMARAN	81

```
In [68]: plt.barh(df["Name"],df["Percentage"])
plt.title("Comparison of HSC Percentage")
plt.xlabel("Name")
plt.ylabel("Exam Percentage")
plt.show()
```



```
In [69]: import pandas as pd
import matplotlib.pyplot as plt
data={
    'Candidate':['Candidate 1', 'Candidate 2', 'Candidate 3', 'Candidate 4'],
    'Votes': [315,130,245,210]
}
df=pd.DataFrame(data)
df.groupby('Candidate').sum().plot(kind='pie',y='Votes')
plt.title("Election Results")
plt.show()
```



```
In [80]: import nltk
from nltk.tokenize import word_tokenize
from nltk.corpus import gutenberg
nltk.download('gutenberg')
nltk.download('punkt')
sample = gutenberg.raw("austen-emma.txt")
token = word_tokenize(sample)
wlist = []
for i in range(50):
    wlist.append(token[i])
wordfreq = [wlist.count(w) for w in wlist]
print("Pairs\n" + str(list(zip(wlist, wordfreq))))
```

```
[nltk_data] Downloading package gutenberg to
[nltk_data] C:\Users\Lenovo\AppData\Roaming\nltk_data...
[nltk_data] Package gutenberg is already up-to-date!
[nltk_data] Downloading package punkt to
[nltk_data] C:\Users\Lenovo\AppData\Roaming\nltk_data...
[nltk_data] Package punkt is already up-to-date!
```

Pairs

```
[(['', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1),
(''], 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2), ('Woo
dhouse', 1), (',', 5), ('handsome', 1), (',', 5), ('clever', 1), (',', 5), ('an
d', 3), ('rich', 1), (',', 5), ('with', 2), ('a', 1), ('comfortable', 1), ('hom
e', 1), ('and', 3), ('happy', 1), ('disposition', 1), (',', 5), ('seemed', 1),
('to', 1), ('unite', 1), ('some', 1), ('of', 2), ('the', 2), ('best', 1), ('ble
ssings', 1), ('of', 2), ('existence', 1), (';', 1), ('and', 3), ('had', 1), ('l
ived', 1), ('nearly', 1), ('twenty-one', 1), ('years', 1), ('in', 1), ('the',
2), ('world', 1), ('with', 2)]
```